

A Multi-Agent Orchestration Framework for Venture Capital Due Diligence

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Abstract

We present a fully automated multi-agent framework for corporate due diligence and market analysis in venture capital. The system runs on an event-driven orchestration architecture, combining Large Language Models (LLMs) with real-time web retrieval to synthesize unstructured data into structured investment intelligence. A central technical contribution is a programmatic extraction pipeline that reverse-engineers the frontend-to-backend communication of the Greek Business Registry (T.E.MH.), querying dynamic endpoints to retrieve official financial filings that are then parsed using a layout-aware OCR extractor. A structural fallback mechanism explicitly flags data absence rather than generating unverified figures, directly targeting hallucination in financial contexts. All workflow artifacts are publicly available to support replication.

Keywords: AI agents; Multi-Agent Systems; Workflow Automation; Due Diligence; Venture Capital; Retrieval-Augmented Generation; Hallucination Mitigation

1 Introduction

15 Evaluating prospective investments and monitoring portfolio companies demands the synthesis
of fragmented information across incompatible sources: unstructured web data, real-time
news, competitive analyses, and official financial registries that are often locked behind
administrative access controls. Manual desk research, the standard approach, introduces
human error, cognitive bias, and delays that compound across a portfolio (Gompers et al.,
20 2016; Kahneman, 2011).

LLMs offer powerful capabilities for natural language understanding and data synthesis
(Bommasani et al., 2021), but deploying them for financial due diligence exposes three
structural problems. First, training data cut-offs leave models blind to current market signals.
Second, LLMs generate plausible but factually incorrect information—a failure mode that
25 financial decision-making cannot tolerate (Ji et al., 2023). Third, base LLMs lack the ability
to autonomously query external databases, invoke APIs, or parse scanned financial documents;
frameworks that enable tool use and inter-agent communication are required to bridge this
gap (Q. Wu et al., 2023).

We address these problems with a fully automated multi-agent orchestration pipeline.
30 Rather than a single LLM prompt, the system decomposes due diligence into specialized
sub-tasks handled by distinct AI agents (Xi et al., 2023). Real-time search APIs supply
current market intelligence; a custom extraction module programmatically retrieves and
parses official financial filings from the Greek Business Registry (Γ.Ε.ΜΗ.).

This paper makes three contributions:

- 35 1. A modular, multi-agent architecture that automates end-to-end corporate research from
a single analyst trigger to a structured HTML report delivered by email.
2. A programmatic financial extraction pipeline that queries hidden Γ.Ε.ΜΗ. registry
endpoints, retrieves raw PDF documents, and applies layout-aware OCR to structure
financial data with preserved source citations.

3. A structural fallback mechanism that routes to open-source financial intelligence when registry data is unavailable, replacing potential LLM hallucinations with explicit, auditable data gaps. 40

2 Background and Related Work

2.1 Multi-Agent Frameworks

The field has evolved from single-prompt LLM interactions toward agentic workflows, where multiple LLM-powered agents each carry distinct roles, system prompts, and tool-use capabilities (Xi et al., 2023). Chaining agents allows each one to pass its verified output to the next, constructing reasoning chains that exceed the capacity of a single model call. Multi-step analytical problems requiring precision and iterative correction are the natural application domain for these architectures. 45 50

AutoGen operationalizes this paradigm through a conversational multi-agent framework in which agents negotiate task decomposition and tool invocation dynamically (Q. Wu et al., 2023). MetaGPT takes a complementary approach, assigning software-engineering roles—product manager, architect, engineer—to distinct agents coordinated via structured output schemas (Hong et al., 2024). LangGraph provides a graph-based state-machine abstraction for agent orchestration, enabling conditional branching and cycle detection in complex pipelines (LangChain Inc., 2024). Our system follows the same DAG-structured philosophy but is implemented on n8n (n8n, 2024), a low-code event-driven platform that exposes HTTP, conditional routing, and AI nodes without requiring custom agent code, substantially lowering the engineering barrier for non-technical investment teams. 55 60

2.2 LLMs for Financial Analysis

Applying agentic systems to finance is natural: general-purpose LLMs struggle with domain-specific vocabulary and numerical precision (Yang et al., 2023). BloombergGPT demonstrated

that domain-adapted pre-training yields measurable improvements on financial NLP benchmarks (S. Wu, Irsoy, et al., 2023). More recent multi-agent architectures have targeted specific financial tasks: MARAG-Fin distributes analytical subtasks across agents to improve trading decision accuracy and reduce hallucinations in volatile markets (Luckianto & Gunawan, 2026), while QuantAgents structures investment analysis as a simulated trading environment coordinated by specialist agents (Li et al., 2025). FinAgent extends this line by incorporating multimodal inputs—price charts, news sentiment, and fundamental data—into a unified agent pipeline (Zhang et al., 2024). Our work is distinguished from this body of literature by its focus on pre-investment due diligence rather than post-investment trading, and by its integration with an official national corporate registry that provides auditable, ground-truth financial data rather than market signals.

2.3 Retrieval-Augmented Generation and Real-Time Grounding

Dynamic information retrieval addresses the static knowledge problem: grounding model outputs in live search results and external data streams allows systems to bypass training cutoffs entirely (Lewis et al., 2020). For corporate intelligence, Open-Source Intelligence (OSINT) provides current market sizing, funding rounds, and competitor movements. Retrieval-Augmented Generation makes that intelligence auditable by linking outputs to primary sources (Gao et al., 2023).

2.4 Document Understanding and OCR

Automated document processing extends retrieval into non-digitized corporate records. Layout-aware parsers handle balance sheets and income statements without the structural loss typical of legacy OCR engines (Paruchuri, 2025). Combined with real-time web retrieval, these tools provide a path from fragmented market signals to formal regulatory filings within a single pipeline—the integration this work demonstrates.

3 System Architecture

The system runs on an event-driven automation platform (n8n, 2024), structured as a Directed Acyclic Graph (DAG) of processing nodes. A custom HTML form serves as the entry point 90 for investment teams: analysts select a target portfolio company and trigger the full research workflow with a single click, with the low-code backend hidden from view.

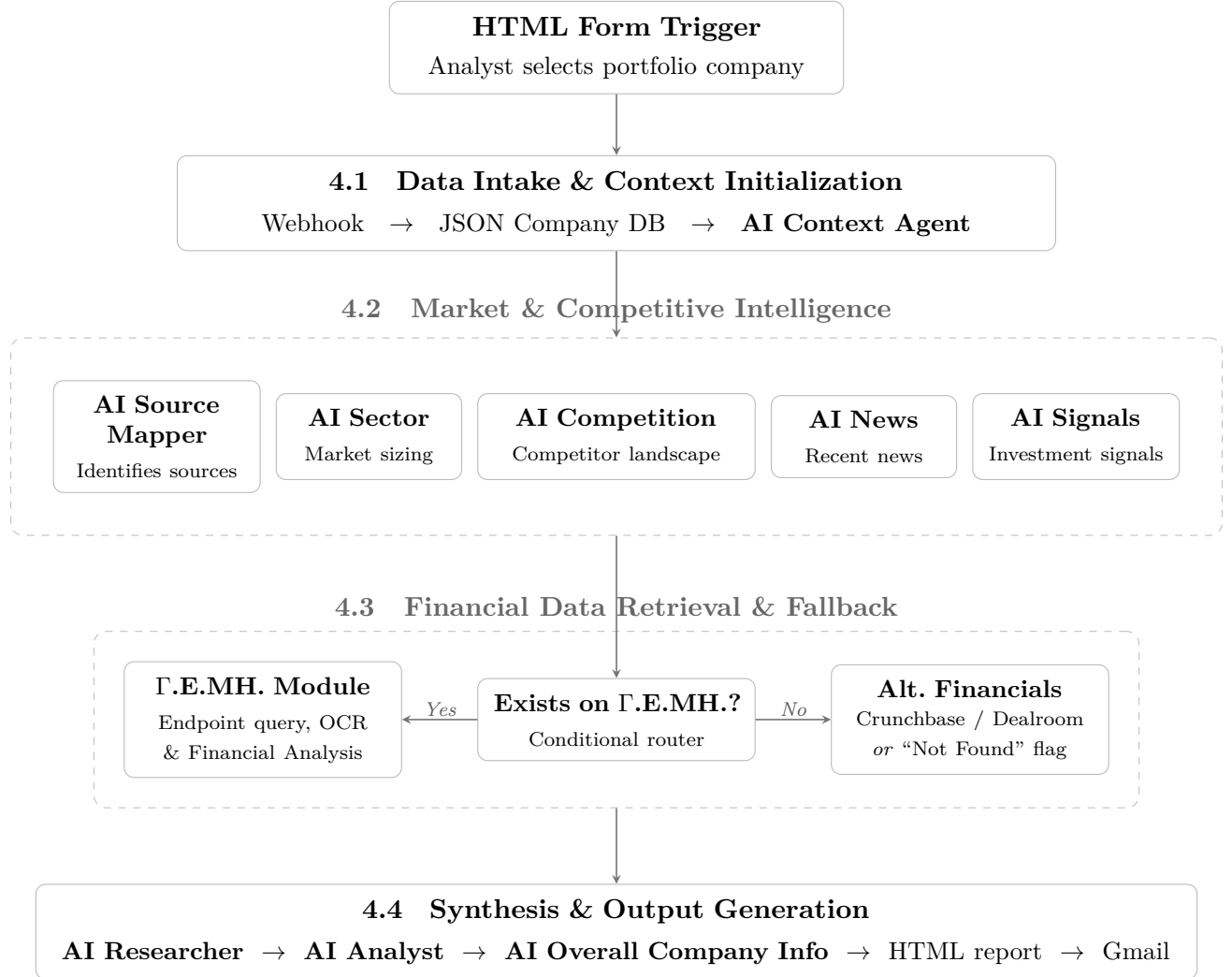


Figure 1: High-level overview of the event-driven multi-agent orchestration architecture, structured according to the four pipeline stages described in Sections 3–3.4.

3.1 Data Intake and Context Initialization

On submission, a Webhook node captures the payload. A transformation node maps the
95 selected company to a pre-scraped JSON database of baseline attributes: founders, sector, initial investment year, headquarters, and registration numbers. The **AI Context Agent** takes this structured input and produces a compact company profile that anchors all downstream research nodes.

3.2 Market and Competitive Intelligence Modules

100 Five specialized agents query the Perplexity Sonar Deep Research API (Perplexity AI, 2024) for real-time market intelligence. The **AI Source Mapper** runs first, identifying the most reliable sector-specific portals, databases, and news aggregators for the target company. The **AI Sector** and **AI Competition** agents then run in parallel against those sources, extracting market sizing, macro-environmental trends, and a competitive landscape that categorizes
105 competitors as direct, adjacent, or niche innovators. Once both complete, the **AI News** and **AI Signals** agents scan for recent strategic developments (partnerships, product launches) and extract quantifiable investment signals.

3.3 Financial Data Retrieval and Fallback Mechanisms

The most technically involved component interfaces with the Greek Business Registry
110 (Γ.Ε.ΜΗ.). Although the registry provides an official API, obtaining authenticated credentials involves administrative overhead. The framework instead reverse-engineers the portal’s frontend-to-backend communication, a method validated by prior open-source work targeting the same registry (Drakakis, 2025).

Registry data and corporate announcements are served as PDF documents loaded from
115 specific backend endpoints. A conditional router checks for a valid registry number (*arGEMI*). When one exists, an HTTP Request node mimics browser behavior, querying those endpoints

to retrieve the JSON payload of document IDs.

Document IDs are classified into two streams: **Corporate Modifications**—covering board changes, capital increases, and statutory modifications—and **Financial Statements**, comprising official balance sheets, income statements, and auditor reports. The pipeline 120 fetches raw PDFs, prioritizing the most recent fiscal years, and passes them through a layout-aware text extractor (Paruchuri, 2025). Two summarization agents, **AI FinSummary** and **AI ModSummary**, distill the extracted text into standardized financial metrics (Assets, Liabilities, Revenue, EBIT) with source citations preserved.

Preventing fabricated financial data is the system’s hardest constraint. When no Γ.E.MH.125 number is found—typically for companies incorporated abroad—the pipeline activates the **Alternative Financials** agent, which queries Crunchbase, Dealroom, and equivalent open-source databases. If those sources also return nothing, the agent writes a “Not Found” flag directly into the report, making the data gap explicit rather than filling it with generated figures. 130

3.4 Synthesis and Output Generation

Three synthesis agents combine the intelligence streams. The **AI Researcher** aggregates sector, competition, news, signals, and financial data into a strategic research note that surfaces recent developments and blind spots. The **AI Analyst** takes that note, produces an Executive Summary, assigns attractiveness scores across market timing and product 135 differentiation, and drafts 30-to-180-day recommendations for both the fund and the startup. The **AI Overall Company Info** agent generates a public-facing summary that founders can use to understand their external market footprint.

A formatting node converts the JSON outputs into a styled HTML report and delivers it to the requesting analyst through a Gmail API integration. 140

4 System Demonstration

To illustrate end-to-end pipeline behavior, we trace a representative execution for a Greek-incorporated FinTech startup registered on Γ.E.MH.

4.1 Trigger and Context Initialization

145 An analyst opens the HTML front-end, selects the target company from the portfolio list, and submits the form. The Webhook node captures the payload and routes it to the transformation node, which retrieves baseline company attributes from the JSON database—sector classification, founding year, headquarters, and Γ.E.MH. registration number (*arGEMI*). The **AI Context Agent** synthesizes these attributes into a compact structured profile that
150 anchors all downstream reasoning nodes to a consistent factual baseline before any external API is queried.

4.2 Parallel Market Intelligence Retrieval

The **AI Source Mapper** identifies the highest-signal information sources for the FinTech sector—specialist databases, regulatory portals, and news aggregators—and propagates them
155 to the five parallel agents. Working concurrently via the Perplexity Sonar Deep Research API (Perplexity AI, 2024), the **AI Sector** agent returns current market sizing estimates and macro-regulatory trends; the **AI Competition** agent identifies and classifies competitors as direct, adjacent, or niche innovators; the **AI News** agent surfaces recent strategic events such as funding announcements and product launches; the **AI Signals** agent extracts quantifiable
160 investment indicators. The merged outputs of all five agents form the market intelligence payload passed to the financial retrieval stage.

4.3 Financial Data Retrieval from Γ .E.MH.

The conditional router detects a valid *arGEMI* and directs execution to the Γ .E.MH. module. An HTTP Request node queries the registry’s backend endpoint, returning a JSON manifest of available document IDs. The classifier separates **Financial Statements**—balance sheets 165 and income statements across available fiscal years—from **Corporate Modifications** such as capital increases and board resolutions. The pipeline fetches the corresponding PDFs, which are processed by the layout-aware OCR extractor (Paruchuri, 2025). The **AI FinSummary** agent structures the extracted text into four standardized line items per fiscal year (Revenue, Total Assets, Total Liabilities, EBIT), preserving direct citations to source document and 170 page number. The **AI ModSummary** agent produces a chronological timeline of structural corporate events. Because all figures originate from official regulatory filings, no generative estimation is required at any stage.

4.4 Synthesis and Report Delivery

The three synthesis agents receive the merged intelligence streams. The **AI Researcher** 175 integrates market, competitive, news, and financial signals into a strategic research note that surfaces recent developments alongside identified analytical blind spots. The **AI Analyst** converts this note into an Executive Summary, assigns attractiveness scores across market timing and product differentiation dimensions, and drafts 30-to-180-day action recommendations for both the fund and the portfolio company. The **AI Overall Company Info** 180 agent generates a founder-facing summary of the company’s external market position. A formatting node assembles the JSON outputs into a styled HTML report, which the Gmail integration delivers directly to the requesting analyst—completing the full pipeline from form submission to delivered report without any intermediate human action required.

185 **5 Limitations**

Several constraints bound the current system. First, the financial extraction component is specific to Γ.E.MH., limiting registry-grade data to Greek-incorporated entities; companies registered abroad rely on open-source databases with lower coverage and less structured data. Second, the pipeline depends on three external commercial services—Perplexity Sonar, 190 the LLM provider, and the Marker PDF extractor—making it subject to pricing changes, rate limits, and interface updates; a production deployment should consider self-hosted alternatives for mission-critical components. Third, the current deployment has been applied within a single Greek venture capital fund; generalizability across investment stages, sectors, and geographies remains to be established through broader empirical evaluation. Fourth, 195 LLM inference is non-deterministic: repeated runs on identical inputs may produce different synthesis outputs, and the variability of the generated sections is currently unquantified.

6 Data and Workflow Availability

All system artifacts are publicly available in a dedicated repository (Alexandrou, 2026): the n8n JSON workflow file, project documentation, and an explanatory video walk-through. 200 Consult the repository documentation for the required environment variables and node configurations before attempting replication.

7 Conclusions and Future Work

Multi-agent orchestration can automate corporate due diligence at a level of reliability previously requiring human analysts. By combining dynamic web retrieval, reverse-engineered 205 endpoint querying, and OCR-based document processing, the pipeline connects unstructured market signals to official financial compliance data in a single automated pass. Structural fallback mechanisms prevent the system from generating unverified financial figures, replacing

potential hallucinations with explicit data gaps.

Three directions extend this work. First, integrating the pipeline with the fund’s internal portfolio database would let each report draw on proprietary deal-level data, historical performance metrics, and analyst notes specific to each startup, making the output contextual rather than generic. Second, adding international registries—such as UK Companies House (UK Companies House, 2025) and equivalent European authorities—would extend financial extraction coverage to companies incorporated outside Greece. Third, replacing the third-party PDF extraction service with a self-hosted open-source alternative would reduce operational costs and remove the external dependency without degrading extraction quality.

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